Explanation of Summary Statistics for Travel Time Differences

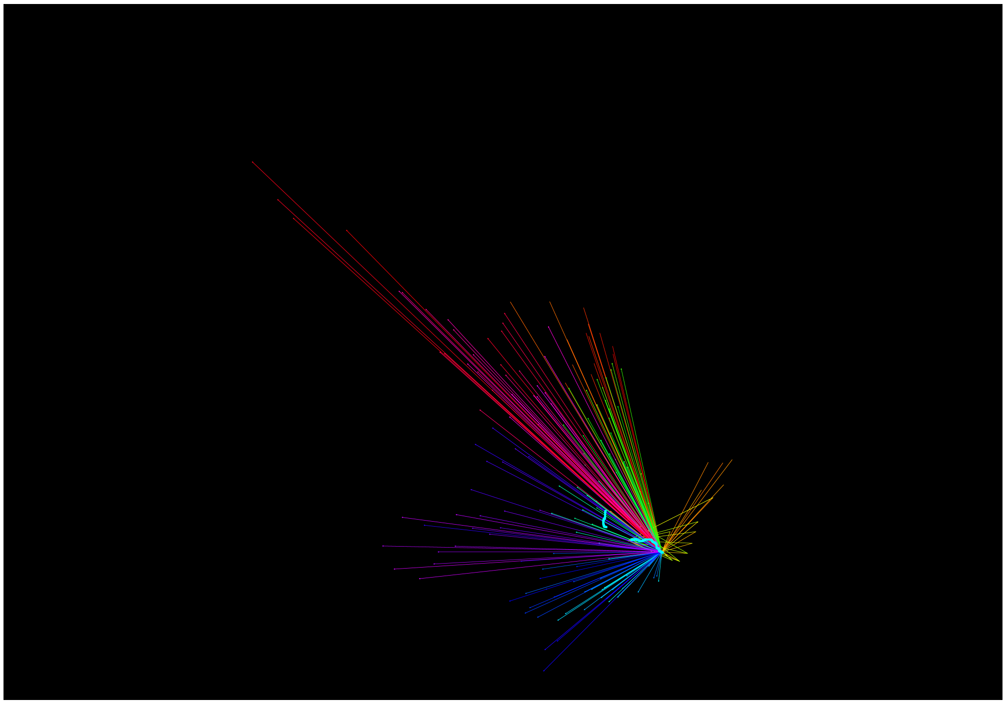
In principle, roads are constructed progressively year by year, so commuting times in later years are usually shorter than in earlier years. However, for rail and High-Speed Rail (HSR), we observe that in some years the travel time between certain city pairs increases relative to the previous year. We analyze these cases one by one. Throughout, we define the change as $\Delta t = t_{\text{previous year}} - t_{\text{current year}}$, so a negative value indicates an increase in travel time.

1 Rail Passengers

1.1 2014-2015

Figure 1 shows the negative change in rail passenger travel times. Bold cyan lines highlight rail segments where travel time is longer in 2015 than in 2014, and arrows indicate city pairs whose travel time is longer in 2015 than in 2014. After checking, we find that the two negative lines are the Ganlong Railway (赣龙铁路) and the Licha Railway (醴茶铁路). This pattern is consistent with passenger services being discontinued in 2015. The negative city pairs are located on either side of these rail segments, suggesting that the observed increases in inter-city travel time are driven by the deterioration along these lines.

Figure 1: rail_hsr: Rail Passengers 2014-2015 Difference

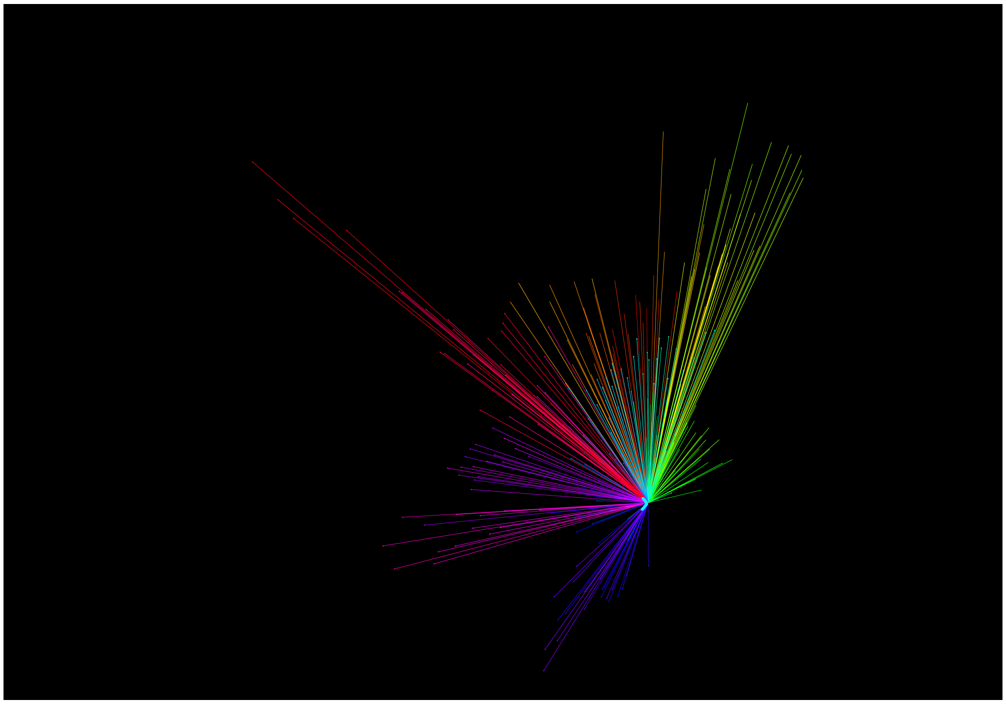


Notes: This figure shows the negative change in rail passenger travel times. Bold cyan lines highlight rail segments where travel time is longer in 2015 than in 2014, and arrows indicate city pairs whose travel time is longer in 2015 than in 2014.

1.2 2012-2013

Similarly, Figure 2 shows that one rail line—Xiangle Railway (向乐铁路)—has a longer travel time in 2013 than in 2012. This pattern is consistent with the service being discontinued in 2013.

Figure 2: rail_hsr: Rail Passengers 2012-2013 Difference

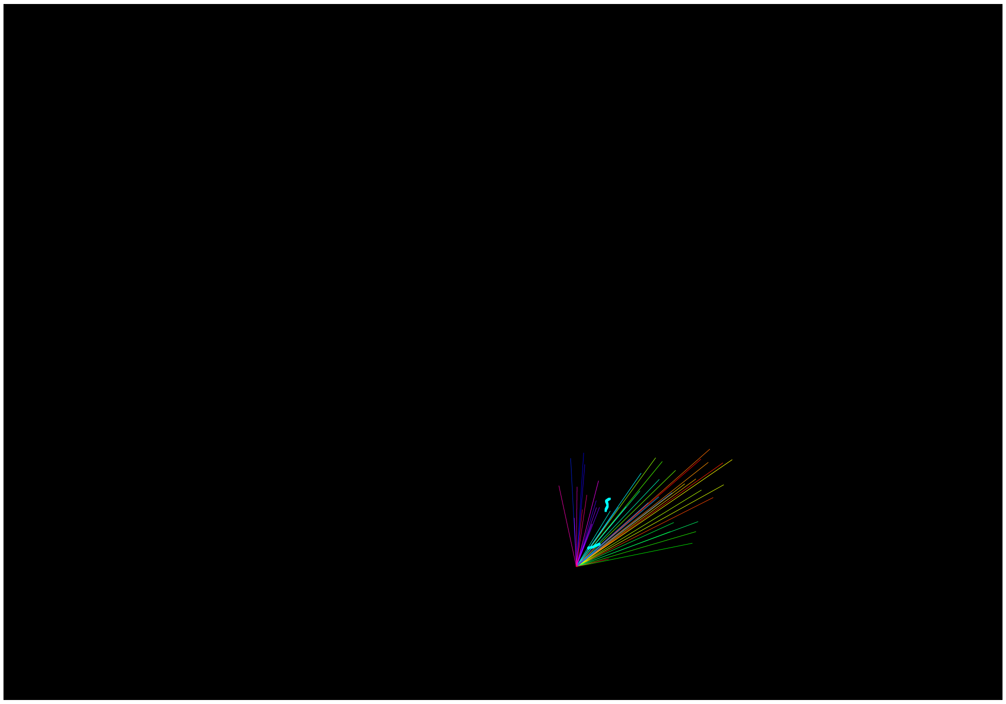


Notes: This figure shows the negative change in rail passenger travel times. Bold cyan lines highlight rail segments where travel time is longer in 2013 than in 2012, and arrows indicate city pairs whose travel time is longer in 2013 than in 2012.

1.3 2003-2004

Figure 3 shows that two rail lines—Chenjia Railway (郴嘉铁路) and Liliu Railway (醴浏铁路)—have longer travel times in 2004 than in 2003. This pattern is consistent with the services on these lines being discontinued in 2004.

Figure 3: rail_hsr: Rail Passengers 2003-2004 Difference, Input

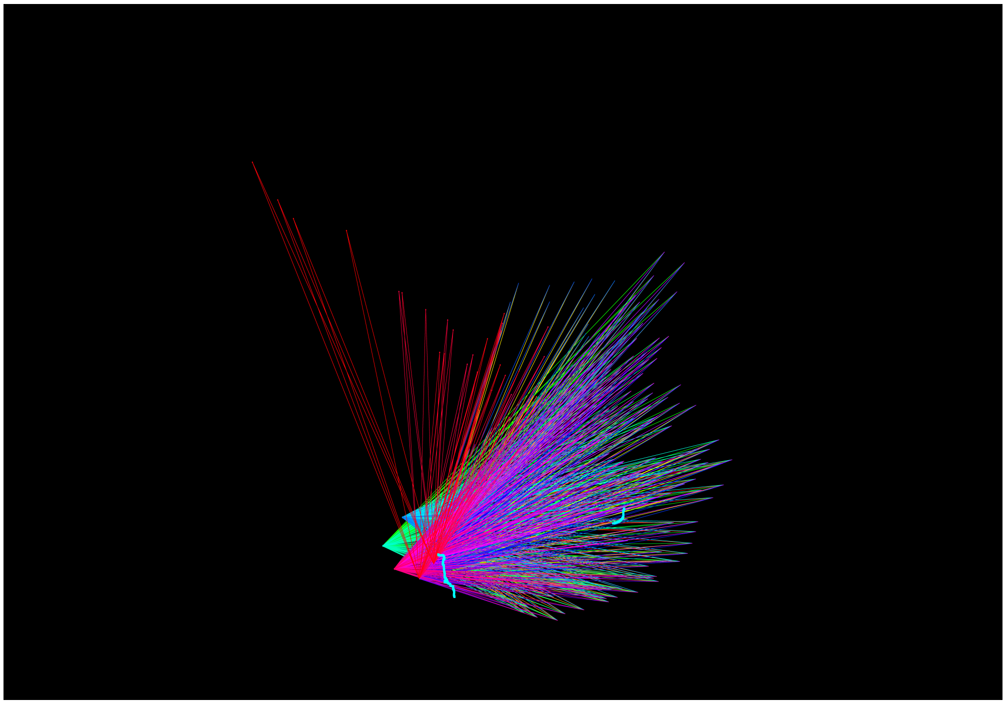


Notes: This figure shows the negative change in rail passenger travel times. Bold cyan lines highlight rail segments where travel time is longer in 2004 than in 2003, and arrows indicate city pairs whose travel time is longer in 2004 than in 2003.

1.4 2002-2003

Figure 4 shows that one rail line—Kunhe Railway (昆河铁路)—has a longer travel time in 2003 than in 2002. This pattern is consistent with passenger service being discontinued in 2003.

Figure 4: rail_hsr: Rail Passengers 2002-2003 Difference



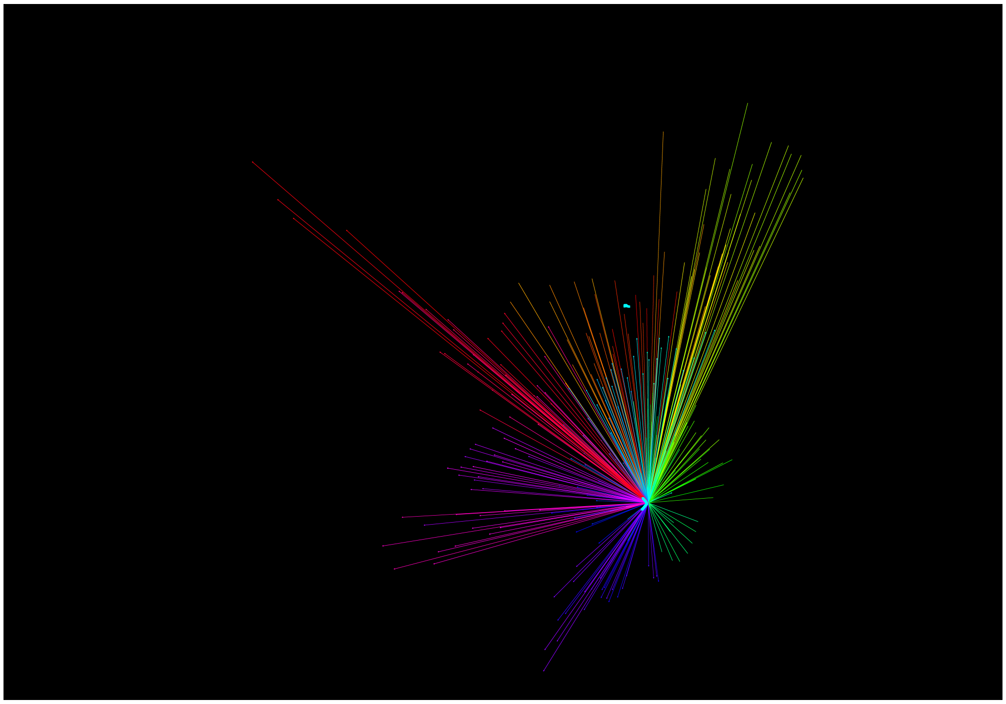
Notes: This figure shows the negative change in rail passenger travel times. Bold cyan lines highlight rail segments where travel time is longer in 2003 than in 2002, and arrows indicate city pairs whose travel time is longer in 2003 than in 2002.

2 Rail Goods

2.1 2012-2013

Figure 5 shows that two rail lines—Xiangle Railway (向乐铁路) and Gaoyi Railway (高易铁路)—have longer travel times in 2013 than in 2012. This pattern is consistent with these lines being discontinued in 2013. Note that the Gaoyi Railway has been goods-only from the outset, so it does not appear in cyan in Figure 2.

Figure 5: rail_hsr: Rail Goods 2012-2013 Difference

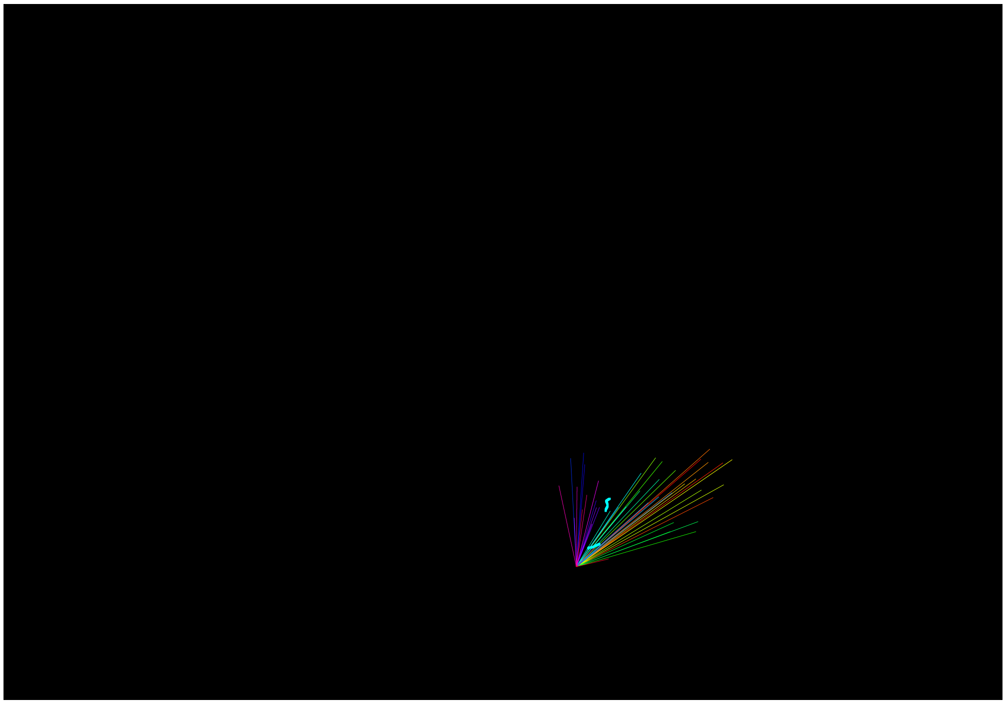


Notes: This figure shows the negative change in rail goods travel times. Bold cyan lines highlight rail segments where travel time is longer in 2013 than in 2012, and arrows indicate city pairs whose travel time is longer in 2013 than in 2012.

2.2 2003-2004

Figure 6 shows that two rail lines—Chenjia Railway (郴嘉铁路) and Liliu Railway (醴浏铁路)—have longer travel times in 2004 than in 2003. This pattern is consistent with these services being discontinued in 2004.

Figure 6: rail_hsr: Rail Goods 2003-2004 Difference



Notes: This figure shows the negative change in rail goods travel times. Bold cyan lines highlight rail segments where travel time is longer in 2004 than in 2003, and arrows indicate city pairs whose travel time is longer in 2004 than in 2003.